Supplementary Materials

S1: Parallel Trends Tests

Each of the following subsections tests one of the parallel trends assumptions PT1 to PT3 for log NO_2, and states the model and sample it uses. Inference throughout uses the wild cluster bootstrap by site (Rademacher weights, 9,999 replications).

S1.1 Linear differential pre-trends (PT1)

The sample covers the pre-period, Jun 2021 to Jul 2022 (14 months), at all 64 monitoring sites. Event time -14 is Jun 2021 and event time -1 is Jul 2022. Each LTN area is compared with the Control Group that model M1 uses for it (see the Estimation section of the main text): BDR Controls for BDR, and the pooled non-BDR Controls otherwise. The regression uses the fixed effects of M1 and adds, for each LTN area r, the regressor $1\{\text{LTN in area } r\} \times$ event time. Its coefficient is the differential drift per month relative to the Control Group. The null hypothesis is that it equals zero.

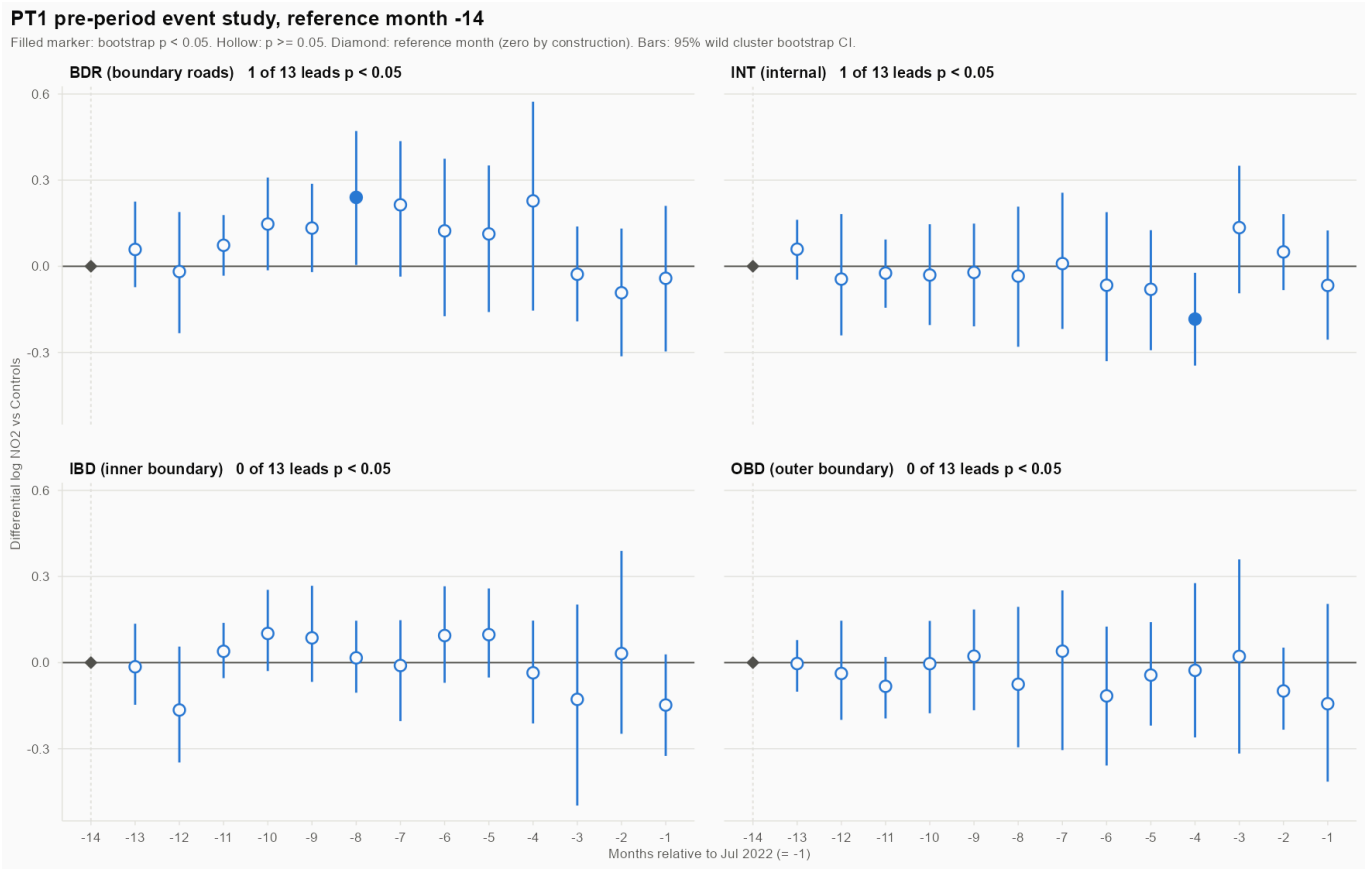
Table S1.1: Differential linear pre-trend in log NO_2 (per month) of each LTN area relative to its Control Group.

Area	Estimate	95% CI	Bootstrap p
BDR	-0.003	[-0.014, 0.008]	0.550
INT	-0.002	[-0.012, 0.008]	0.659
IBD	-0.003	[-0.014, 0.007]	0.491
OBD	-0.006	[-0.018, 0.011]	0.397

No area shows a significant differential trend. The point estimates are small, at most about 0.6% per month in absolute value, and the confidence intervals include zero. We therefore do not reject the hypothesis that, in each area, the Treated and Control Groups follow linear trends with the same slope before the LTN schemes were implemented.

S1.2 Event study (PT1)

To look for departures from a linear trend, we use the sample and fixed effects of S1.1 but replace the linear trend with one lead coefficient per pre-period month, relative to the first pre-period month (Jun 2021, event time -14). Figure S1 shows the 13 leads for each area.



Two of the 52 leads are significant at the 5% level: one in BDR (event time -8) and one in INT (event time -4). With 52 tests, about 2.6 rejections are expected by chance alone. The estimates show no systematic drift as treatment approaches.

With the last pre-period month (-1) as the reference, 8 of the 52 leads are significant, 6 of them in IBD. This is because Jul 2022 is unusually low for IBD LTN sites, which shifts every lead when it is used as the baseline. We therefore use the first pre-period month, which is not affected by this single-month deviation.

S1.3 Differences in pre-trends between areas (PT2)

We begin by rearranging PT2:

$$\theta_a - \phi_a = \theta_b - \phi_b$$

In other words, the Treated Group's trend in the absence of the LTN schemes, relative to its Control Group, is the same in areas a and b. We test this before the LTN schemes were implemented.

Let ψ_k denote the slope estimated for area k in S1.1. It is the monthly drift of the Treated Group relative to its Control Group before the schemes, and so the pre-period counterpart of $\theta_k - \phi_k$. The pre-period analogue of PT2 is therefore $\psi_a = \psi_b$. For each pair of areas, the null hypothesis is $\psi_a - \psi_b = 0$. The contrasts are estimated from the model and sample of S1.1, which include all four areas.

Table S1.2: Difference in differential linear pre-trends between pairs of LTN areas (per month).

Contrast	Estimate	95% CI	Bootstrap p
BDR − INT	-0.001	[-0.015, 0.013]	0.874
BDR − IBD	0.000	[-0.015, 0.015]	0.976

Contrast	Estimate	95% CI	Bootstrap p
BDR − OBD	0.002	[-0.015, 0.019]	0.776
INT − IBD	0.001	[-0.009, 0.011]	0.778
INT − OBD	0.003	[-0.012, 0.016]	0.589
IBD − OBD	0.002	[-0.013, 0.015]	0.755

None of the contrasts is significant, and all are close to zero. We therefore do not reject the null hypothesis that, in any two areas, the Treated Groups' trends relative to their Control Groups are the same before the LTN schemes were implemented.

S1.4 Parallel Control Group trends across non-BDR areas (PT3)

Control Group trends are observed in both policy periods. The sample therefore covers the non-BDR Control sites (11 in INT and 9 in IBD areas) over the full period, Jun 2021 to Dec 2023, excluding the rollout window (Aug to Oct 2022). There are no Control sites in OBD areas (see the Data section of the main text), so ϕ_{OBD} cannot be estimated. The testable part of PT3 is therefore $\phi_{\mathrm{INT}} = \phi_{\mathrm{IBD}}$, with the series subscript suppressed. We estimate

$$Y_{i,j} = \alpha_i + \beta_j + \kappa\, 1\{T_j = Post\}\, 1\{A_i = \mathrm{IBD}\} + \epsilon_{i,j}$$

where α_i are site fixed effects and β_j are time fixed effects. The coefficient κ is the difference between the Control Group trends, $\phi_{\mathrm{IBD}} - \phi_{\mathrm{INT}}$, on the log scale. The null hypothesis is $\kappa = 0$.

Table S1.3: Difference in Control Group trends in log NO_2 between IBD and INT areas, Post vs Pre.

Contrast	Estimate	95% CI	Bootstrap p
IBD − INT	-0.011	[-0.093, 0.062]	0.794

The difference is small and not significant, so we do not reject PT3. With only 20 Control sites, however, the test has limited power: the confidence interval is consistent with differences between about −9% and +6%.

S2: Derivations for Estimation and Differential Treatment Effects

Unless stated otherwise, the series s is held fixed and suppressed, so that subscripts such as INT refer to the area-series pair (INT, s).

S2.1 The OLS estimand of M1

M1 has separate time fixed effects for BDR and non-BDR sites, so its OLS normal equations split into two independent blocks. Consider one block. Let L_r denote the LTN Treated sites in area r, and C the Control sites in the block. In the non-BDR block, C pools the Control sites of all non-BDR areas. Let $\bar{y}_S^T$ denote the sample mean over sites in S and times in policy period T, and $\bar{\hat{\beta}}^T$ the average estimated time fixed effect in period T. Assume for now that the panel is balanced, i.e., every site in the block is observed at every time.

The normal equations state that the OLS residuals sum to zero over each site (for α_i), over each time (for β_j), and over the post-period observations of L_r (for δ_r). We use them in three steps.

1. For each site in L_r, the residuals sum to zero over all times. Combined with the equation for δ_r, the residuals of L_r sum to zero within each policy period. Site fixed effects cancel, because each site contributes equally to the means of both periods. Hence

$$\bar{y}_{L_r}^{Post} - \bar{y}_{L_r}^{Pre} = \left(\bar{\hat{\beta}}^{Post} - \bar{\hat{\beta}}^{Pre}\right) + \hat{\delta}_r$$

2. Summing the equations for β_j over the times in each policy period, the residuals of all sites in the block sum to zero within each period. By step 1, the LTN Treated sites contribute zero, so the residuals of C also sum to zero within each period. Hence

$$\bar{y}_{C}^{Post} - \bar{y}_{C}^{Pre} = \bar{\hat{\beta}}^{Post} - \bar{\hat{\beta}}^{Pre}$$

3. Subtracting step 2 from step 1,

$$\hat{\delta}_r = \left(\bar{y}_{L_r}^{Post} - \bar{y}_{L_r}^{Pre}\right) - \left(\bar{y}_{C}^{Post} - \bar{y}_{C}^{Pre}\right)$$

In the notation of the main text, $\hat{\delta}^k_{OLS} = \hat{\Delta}_k - \hat{c}_k$, where $\hat{c}_k = \hat{\phi}_k$ for BDR areas and $\hat{c}_k = \hat{\phi}_{\text{NBDR}}$ for non-BDR areas. Taking expectations, $\hat{\delta}^k_{OLS}$ is unbiased for $\Delta_k - c_k$, where c_k is the Control Group trend that M1 uses for k. For BDR areas, $c_k = \phi_k$, giving τ^k_{DiD}. For non-BDR areas, $c_k = \phi_{\text{NBDR}}$, giving $\tau^k_{\text{DiD,NBDR}}$.

This result is purely algebraic. It relies neither on the dependence structure of M1, which is needed only for inference, nor on any parallel trends assumption. It therefore still holds when PT1 to PT4 fail, as S2.2 and S2.3 require. Our panels are not fully balanced, since some sites are not observed at every time. The equality in step 3 then holds only approximately.

S2.2 Differential treatment effects in general

From the definition of the treatment effect,

$$\gamma_{a,b} = \tau_a - \tau_b = (\Delta_a - \theta_a) - (\Delta_b - \theta_b) = (\Delta_a - \Delta_b) - (\theta_a - \theta_b)$$

By S2.1, $\hat{\gamma}^{a,b}_{OLS} = \hat{\delta}^a_{OLS} - \hat{\delta}^b_{OLS}$ is unbiased for $(\Delta_a - c_a) - (\Delta_b - c_b)$. Hence

$$\mathbb{E}\left[\hat{\gamma}^{a,b}_{OLS}\right] - \gamma_{a,b} = (\theta_a - \theta_b) - (c_a - c_b)$$

which is zero if and only if $\theta_a - \theta_b = c_a - c_b$. The table below summarises this condition for each type of pair. Here s and s' denote the series of a and b, and r, r' denote non-BDR areas.

a **vs** b	**Estimand of** $\hat{\gamma}^{a,b}_{OLS}$	**Necessary and sufficient condition**	**Sufficient assumptions**
BDR vs BDR, different series	$(\Delta_a - \Delta_b) - (\phi_a - \phi_b)$	$\theta_a - \theta_b = \phi_a - \phi_b$	PT2
BDR vs non-BDR	$(\Delta_a - \Delta_b) - (\phi_a - \phi_{(\text{NBDR},s')})$	$\theta_a - \theta_b = \phi_a - \phi_{(\text{NBDR},s')}$	PT2 and PT3
non-BDR vs non-BDR, different series	$(\Delta_a - \Delta_b) - (\phi_{(\text{NBDR},s)} - \phi_{(\text{NBDR},s')})$	$\theta_a - \theta_b = \phi_{(\text{NBDR},s)} - \phi_{(\text{NBDR},s')}$	PT2 and PT3

a vs b	Estimand of $\hat{\gamma}_{OLS}^{a,b}$	Necessary and sufficient condition	Sufficient assumptions
non-BDR vs non-BDR, same series	$\Delta_a - \Delta_b$	$\theta_{(r,s)} = \theta_{(r',s)}$	PT4

If a and b are both in BDR areas, or if PT3 holds, then $c_k = \phi_k$ for both, $\hat{\gamma}_{OLS}^{a,b}$ is unbiased for $\gamma_{\text{DDD}}^{a,b}$, and

$$\mathbb{E}\left[\hat{\gamma}_{OLS}^{a,b}\right] - \gamma_{a,b} = \gamma_{\text{DDD}}^{a,b} - \gamma_{a,b} = (\theta_a - \phi_a) - (\theta_b - \phi_b)$$

which is zero if and only if PT2 holds. PT2 can hold even when PT1 fails for both a and b. Take heavy and light vehicles in the same area, and suppose that PT1 would hold for both classes but for a shock specific to the LTN Treated Group. If the shock changes the counterfactual trend of both classes by ω on the log scale, i.e., by the same proportion, then $\theta_a - \phi_a = \theta_b - \phi_b = \omega$, and the bias cancels. In non-BDR areas, this argument also relies on PT3.

S2.3 PT4 is strictly weaker than PT2 and PT3 together

PT2 and PT3 together imply PT4. For two non-BDR areas and the same series, PT3 sets $\phi_a = \phi_b$, so PT2 gives $\theta_a = \theta_b$. Since PT1 for both areas implies PT2, PT1 and PT3 together also imply PT4.

The converse does not hold. Suppose that, absent the LTN schemes, the LTN Treated Groups in INT and IBD areas would both have grown by about 2%, while the Control Groups grew by about 2% in INT areas and 5% in IBD areas. On the log scale,

$$\theta_{\text{INT}} = \theta_{\text{IBD}} = 0.02, \qquad \phi_{\text{INT}} = 0.02, \qquad \phi_{\text{IBD}} = 0.05$$

PT4 holds for INT and IBD areas. However, PT3 fails, PT1 fails within IBD areas, and PT2 fails, since $\theta_{\text{INT}} - \theta_{\text{IBD}} = 0 \neq \phi_{\text{INT}} - \phi_{\text{IBD}}$. The pooled Control Group trend ϕ_{NBDR} lies strictly between 0.02 and 0.05. By S2.1, both $\hat{\delta}_{OLS}^{\text{INT}}$ and $\hat{\delta}_{OLS}^{\text{IBD}}$ are therefore biased for the treatment effect, each by $0.02 - \phi_{\text{NBDR}} \neq 0$. Because the two biases are equal, their difference $\hat{\gamma}_{OLS}^{\text{INT,IBD}}$ remains unbiased.

S3: OLS Estimation Results

This section reports every OLS coefficient of M1, including the time fixed effects: Table S3.1 for NO_2 and Table S3.2 for the four traffic volume series, which are estimated jointly. All estimates are on the log scale. Site fixed effects are absorbed and not reported.

For every coefficient, the "Cluster-robust SE" and "Cluster-robust p" columns give the conventional (sandwich) inference, for reference only. Our inference rests on the wild cluster bootstrap (Rademacher weights, 9,999 replications), shown in the "Bootstrap 95% CI" and "Bootstrap p" columns. We ran it for the treatment effects δ only, so these columns are blank for the time fixed effects. Applying $100\% \times (\exp(\cdot) - 1)$ to these estimates and CIs gives Table 3 of the main text.

M1 has a separate set of time fixed effects for each block of sites: BDR and NBDR for NO_2, and each combination of series and B_i for traffic volumes. The site fixed effects absorb the level of each block, so the time fixed effects are identified only relative to a reference level within the block. One time fixed effect per block is therefore set to zero, and the others in the block are measured relative to it. The omitted levels are listed below each table. The within R^2 measures the fit net of the site fixed effects, so it includes the contribution of the time fixed effects.

S3.1 NO_2

Time j is a calendar month, labelled YYYY-MM. The series subscript is suppressed, as in S2.

Table S3.1: OLS estimates of M1 for log NO_2 (upper), with sample and fit statistics (lower).

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
δ_r	BDR	0.042	0.056	0.452	[-0.077, 0.158]	0.471
δ_r	INT	-0.091	0.029	0.002	[-0.153, -0.027]	0.009
δ_r	IBD	0.014	0.028	0.615	[-0.048, 0.078]	0.628
δ_r	OBD	0.043	0.054	0.425	[-0.101, 0.174]	0.504
$\beta_{j,t}$	2021-06, NBDR	-0.323	0.040	<0.001	–	–
$\beta_{j,t}$	2021-07, BDR	0.028	0.033	0.404	–	–
$\beta_{j,t}$	2021-07, NBDR	-0.324	0.039	<0.001	–	–
$\beta_{j,t}$	2021-08, BDR	-0.207	0.037	<0.001	–	–
$\beta_{j,t}$	2021-08, NBDR	-0.520	0.048	<0.001	–	–
$\beta_{j,t}$	2021-09, BDR	0.286	0.032	<0.001	–	–
$\beta_{j,t}$	2021-09, NBDR	-0.007	0.033	0.826	–	–
$\beta_{j,t}$	2021-10, BDR	0.175	0.044	<0.001	–	–
$\beta_{j,t}$	2021-10, NBDR	-0.092	0.037	0.015	–	–
$\beta_{j,t}$	2021-11, BDR	0.230	0.047	<0.001	–	–
$\beta_{j,t}$	2021-11, NBDR	0.028	0.036	0.445	–	–
$\beta_{j,t}$	2021-12, BDR	0.239	0.058	<0.001	–	–
$\beta_{j,t}$	2021-12, NBDR	0.024	0.032	0.466	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,t}$	2022-01, BDR	0.363	0.062	<0.001	–	–
$\beta_{j,t}$	2022-01, NBDR	0.157	0.039	<0.001	–	–
$\beta_{j,t}$	2022-02, BDR	0.208	0.057	<0.001	–	–
$\beta_{j,t}$	2022-02, NBDR	0.047	0.032	0.147	–	–
$\beta_{j,t}$	2022-03, BDR	0.221	0.054	<0.001	–	–
$\beta_{j,t}$	2022-03, NBDR	0.038	0.034	0.274	–	–
$\beta_{j,t}$	2022-04, BDR	-0.297	0.073	<0.001	–	–
$\beta_{j,t}$	2022-04, NBDR	-0.426	0.050	<0.001	–	–
$\beta_{j,t}$	2022-05, BDR	-0.109	0.044	0.017	–	–
$\beta_{j,t}$	2022-05, NBDR	-0.461	0.055	<0.001	–	–
$\beta_{j,t}$	2022-06, BDR	-0.054	0.067	0.428	–	–
$\beta_{j,t}$	2022-06, NBDR	-0.350	0.050	<0.001	–	–
$\beta_{j,t}$	2022-07, BDR	0.072	0.067	0.285	–	–
$\beta_{j,t}$	2022-07, NBDR	-0.203	0.043	<0.001	–	–
$\beta_{j,t}$	2022-11, BDR	0.371	0.072	<0.001	–	–
$\beta_{j,t}$	2022-11, NBDR	0.073	0.041	0.076	–	–
$\beta_{j,t}$	2022-12, BDR	0.246	0.067	<0.001	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,t}$	2022-12, NBDR	0.103	0.031	0.002	–	–
$\beta_{j,t}$	2023-01, BDR	0.214	0.116	0.070	–	–
$\beta_{j,t}$	2023-01, NBDR	0.084	0.036	0.021	–	–
$\beta_{j,t}$	2023-02, BDR	0.387	0.052	<0.001	–	–
$\beta_{j,t}$	2023-02, NBDR	0.191	0.033	<0.001	–	–
$\beta_{j,t}$	2023-03, BDR	0.173	0.044	<0.001	–	–
$\beta_{j,t}$	2023-03, NBDR	-0.050	0.037	0.187	–	–
$\beta_{j,t}$	2023-04, BDR	0.100	0.042	0.021	–	–
$\beta_{j,t}$	2023-04, NBDR	-0.108	0.048	0.028	–	–
$\beta_{j,t}$	2023-05, BDR	0.072	0.055	0.199	–	–
$\beta_{j,t}$	2023-05, NBDR	-0.208	0.062	0.001	–	–
$\beta_{j,t}$	2023-06, BDR	-0.132	0.098	0.184	–	–
$\beta_{j,t}$	2023-06, NBDR	-0.309	0.044	<0.001	–	–
$\beta_{j,t}$	2023-07, BDR	-0.072	0.047	0.132	–	–
$\beta_{j,t}$	2023-07, NBDR	-0.322	0.047	<0.001	–	–
$\beta_{j,t}$	2023-08, BDR	-0.194	0.056	0.001	–	–
$\beta_{j,t}$	2023-08, NBDR	-0.429	0.042	<0.001	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,t}$	2023-09, BDR	0.410	0.098	<0.001	–	–
$\beta_{j,t}$	2023-09, NBDR	-0.032	0.049	0.509	–	–
$\beta_{j,t}$	2023-10, BDR	0.536	0.094	<0.001	–	–
$\beta_{j,t}$	2023-10, NBDR	0.113	0.019	<0.001	–	–
$\beta_{j,t}$	2023-11, BDR	0.394	0.102	<0.001	–	–
$\beta_{j,t}$	2023-11, NBDR	0.115	0.020	<0.001	–	–
$\beta_{j,t}$	2023-12, BDR	0.363	0.091	<0.001	–	–

Statistic	Value
Observations	1,669
Site fixed effects	64
Clusters (sites)	64
Omitted time effects (set to zero)	2021-06, BDR; 2023-12, NBDR
RMSE	0.196
Adjusted R^2	0.708
Within R^2	0.539

S3.2 Traffic volume

Time j is a day of the week within a data collection wave (2021 or 2023). Each series is observed only on its own day type, so the weekday series have time fixed effects for Monday to Friday only, and the weekend series for Saturday and Sunday only.

Table S3.2: OLS estimates of M1 for log daily traffic volume (upper), with sample and fit statistics (lower).

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\delta_{r,s}$	BDR, LV-WKD	-0.007	0.064	0.908	[-0.169, 0.185]	0.939
$\delta_{r,s}$	INT, LV-WKD	-0.754	0.177	<0.001	[-1.231, -0.268]	0.006

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\delta_{r,s}$	IBD, LV-WKD	-0.679	0.103	<0.001	[-0.963, -0.391]	<0.001
$\delta_{r,s}$	OBD, LV-WKD	-0.179	0.164	0.278	[-0.654, 0.267]	0.471
$\delta_{r,s}$	BDR, LV-WKE	-0.020	0.064	0.756	[-0.182, 0.168]	0.847
$\delta_{r,s}$	INT, LV-WKE	-0.649	0.209	0.002	[-1.195, -0.051]	0.035
$\delta_{r,s}$	IBD, LV-WKE	-0.489	0.143	<0.001	[-0.855, -0.042]	0.035
$\delta_{r,s}$	OBD, LV-WKE	-0.067	0.198	0.737	[-0.617, 0.508]	0.822
$\delta_{r,s}$	BDR, HV-WKD	0.195	0.063	0.002	[0.019, 0.369]	0.030
$\delta_{r,s}$	INT, HV-WKD	-0.272	0.172	0.116	[-0.756, 0.194]	0.272
$\delta_{r,s}$	IBD, HV-WKD	-0.238	0.133	0.076	[-0.612, 0.128]	0.212
$\delta_{r,s}$	OBD, HV-WKD	0.302	0.177	0.089	[-0.213, 0.787]	0.253
$\delta_{r,s}$	BDR, HV-WKE	0.099	0.088	0.258	[-0.141, 0.337]	0.429
$\delta_{r,s}$	INT, HV-WKE	-0.434	0.181	0.017	[-0.941, 0.058]	0.085
$\delta_{r,s}$	IBD, HV-WKE	-0.409	0.144	0.005	[-0.819, -0.014]	0.042
$\delta_{r,s}$	OBD, HV-WKE	-0.143	0.219	0.516	[-0.757, 0.462]	0.665
$\beta_{j,s,t}$	2021, Fri, HV-WKD, NBDR	0.204	0.100	0.042	–	–
$\beta_{j,s,t}$	2021, Fri, LV-WKD, BDR	0.114	0.052	0.029	–	–
$\beta_{j,s,t}$	2021, Fri, LV-WKD, NBDR	0.214	0.055	<0.001	–	–
$\beta_{j,s,t}$	2021, Mon, HV-WKD, BDR	0.047	0.025	0.062	–	–
$\beta_{j,s,t}$	2021, Mon, HV-WKD, NBDR	0.071	0.098	0.466	–	–
$\beta_{j,s,t}$	2021, Mon, LV-WKD, BDR	0.015	0.051	0.764	–	–
$\beta_{j,s,t}$	2021, Mon, LV-WKD, NBDR	-0.027	0.054	0.613	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,s,t}$	2021, Sat, HV-WKE, BDR	0.220	0.038	<0.001	–	–
$\beta_{j,s,t}$	2021, Sat, HV-WKE, NBDR	0.178	0.109	0.104	–	–
$\beta_{j,s,t}$	2021, Sat, LV-WKE, BDR	0.164	0.051	0.001	–	–
$\beta_{j,s,t}$	2021, Sat, LV-WKE, NBDR	0.257	0.115	0.026	–	–
$\beta_{j,s,t}$	2021, Sun, HV-WKE, BDR	-0.127	0.036	<0.001	–	–
$\beta_{j,s,t}$	2021, Sun, HV-WKE, NBDR	-0.251	0.110	0.023	–	–
$\beta_{j,s,t}$	2021, Sun, LV-WKE, BDR	0.056	0.049	0.259	–	–
$\beta_{j,s,t}$	2021, Sun, LV-WKE, NBDR	0.062	0.108	0.565	–	–
$\beta_{j,s,t}$	2021, Thu, HV-WKD, BDR	0.054	0.016	0.001	–	–
$\beta_{j,s,t}$	2021, Thu, HV-WKD, NBDR	0.127	0.096	0.189	–	–
$\beta_{j,s,t}$	2021, Thu, LV-WKD, BDR	0.089	0.050	0.076	–	–
$\beta_{j,s,t}$	2021, Thu, LV-WKD, NBDR	0.103	0.052	0.050	–	–
$\beta_{j,s,t}$	2021, Tue, HV-WKD, BDR	0.028	0.025	0.266	–	–
$\beta_{j,s,t}$	2021, Tue, HV-WKD, NBDR	0.085	0.097	0.384	–	–
$\beta_{j,s,t}$	2021, Tue, LV-WKD, BDR	0.035	0.051	0.490	–	–
$\beta_{j,s,t}$	2021, Tue, LV-WKD, NBDR	0.006	0.053	0.913	–	–
$\beta_{j,s,t}$	2021, Wed, HV-WKD, BDR	0.018	0.018	0.307	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,s,t}$	2021, Wed, HV-WKD, NBDR	0.098	0.095	0.303	–	–
$\beta_{j,s,t}$	2021, Wed, LV-WKD, BDR	0.051	0.050	0.308	–	–
$\beta_{j,s,t}$	2021, Wed, LV-WKD, NBDR	0.050	0.053	0.347	–	–
$\beta_{j,s,t}$	2023, Fri, HV-WKD, BDR	-0.012	0.027	0.647	–	–
$\beta_{j,s,t}$	2023, Fri, HV-WKD, NBDR	-0.035	0.027	0.195	–	–
$\beta_{j,s,t}$	2023, Fri, LV-WKD, BDR	0.052	0.013	<0.001	–	–
$\beta_{j,s,t}$	2023, Fri, LV-WKD, NBDR	0.035	0.020	0.080	–	–
$\beta_{j,s,t}$	2023, Mon, HV-WKD, BDR	-0.016	0.031	0.613	–	–
$\beta_{j,s,t}$	2023, Mon, HV-WKD, NBDR	-0.053	0.030	0.081	–	–
$\beta_{j,s,t}$	2023, Mon, LV-WKD, BDR	-0.016	0.014	0.259	–	–
$\beta_{j,s,t}$	2023, Mon, LV-WKD, NBDR	-0.032	0.015	0.034	–	–
$\beta_{j,s,t}$	2023, Sat, HV-WKE, BDR	0.203	0.044	<0.001	–	–
$\beta_{j,s,t}$	2023, Sat, HV-WKE, NBDR	0.216	0.037	<0.001	–	–
$\beta_{j,s,t}$	2023, Sat, LV-WKE, BDR	0.097	0.014	<0.001	–	–
$\beta_{j,s,t}$	2023, Sat, LV-WKE, NBDR	0.093	0.015	<0.001	–	–
$\beta_{j,s,t}$	2023, Thu, HV-WKD, BDR	-0.013	0.029	0.647	–	–
$\beta_{j,s,t}$	2023, Thu, HV-WKD, NBDR	0.002	0.026	0.949	–	–

Parameter	Level	Estimate	Cluster-robust SE	Cluster-robust p	Bootstrap 95% CI	Bootstrap p
$\beta_{j,s,t}$	2023, Thu, LV-WKD, BDR	-0.010	0.011	0.349	–	–
$\beta_{j,s,t}$	2023, Thu, LV-WKD, NBDR	-0.011	0.018	0.560	–	–
$\beta_{j,s,t}$	2023, Tue, HV-WKD, BDR	0.001	0.031	0.969	–	–
$\beta_{j,s,t}$	2023, Tue, HV-WKD, NBDR	0.022	0.032	0.488	–	–
$\beta_{j,s,t}$	2023, Tue, LV-WKD, BDR	-0.011	0.009	0.225	–	–
$\beta_{j,s,t}$	2023, Tue, LV-WKD, NBDR	0.011	0.022	0.630	–	–
$\beta_{j,s,t}$	2023, Wed, HV-WKD, BDR	0.010	0.029	0.739	–	–

Statistic	Value
Observations	3,828
Site × series fixed effects	548
Clusters (site × wave)	274
Omitted time effects (set to zero)	2021, Fri, HV-WKD, BDR; 2023, Sun, HV-WKE, BDR; 2023, Sun, HV-WKE, NBDR; 2023, Sun, LV-WKE, BDR; 2023, Sun, LV-WKE, NBDR; 2023, Wed, HV-WKD, NBDR; 2023, Wed, LV-WKD, BDR; 2023, Wed, LV-WKD, NBDR
RMSE	0.489
Adjusted R^2	0.934
Within R^2	0.219